



here are my results..help me interpret them.

Orchestrated breakdown of experimental findings

Experiment A (capacity):

More capacity → better validation accuracy, and faster convergence too. The tiny network (8 units) tops out around ~75-77% val accuracy after 50 epochs and is still climbing slowly — it's underfitting, lacking enough parameters to capture the pattern in the data. The default (64, 32) reaches ~90% and plateaus by epoch 30. The large network (256,128,64) both converges fastest (near its max by epoch 10) and reaches the highest val accuracy

lets answer this critically but briefly and shortly showing me the criteria we check to get our solution.

Training 1. Map each line of the PyTorch loop to the NumPy code you wrote in Section 1: which line is the forward pass, which computes gradients, which is the update rule? 2. Looking at your curves, is the model overfitting.

Identified overfitting patterns through quantitative metric analysis

1. Mapping PyTorch lines to your Section 1 NumPy code:

PyTorch line

Section 1 equivalent

`logits = model(xb)`

forward pass — `y_hat = w*x + b` (Part 1.1) or the `z1/h/y_hat` chain (Part 1.2)

`loss = criterion(logits, yb)`

`loss = np.mean((y_hat - y)**2)`

i simply re-run all cells

Glad that did it — sometimes it really is just a transient network blip. Confirm you now see the dataframe loaded (e.g., run `obesity.head()` to check it printed rows), and then we're clear to move on to the categorical encoding step whenever you're ready.

i used ordinal encoding for caec and calc then one hot encoding for gender and mtrans

I need two things from your Lab 2 work to keep this consistent with what you already built — I can't guess these without risking mismatched code.

i used ordinal encoding for caec and calc then one hot encoding for gender and mtrans

lets answer this critically but briefly showing what metric is to be used to evaluate the answers. i want to understand.

1. With tanh removed, the deep network fits no better than the single neuron of Part 1.1. Explain mathematically why a stack of linear layers collapses into one linear layer. 2. What role did the hidden layer's 8 neurons play in fitting

Synthesized mathematical foundations and evaluation metrics for neural network concepts

Here's each answer, with the metric to check noted alongside:

1. Why stacked linear layers collapse into one linear layer